

Xiaoyan Cong

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Education

Brown University , Ph.D. Student of Computer Science	September 2024 – Present
• Advisor: <i>Professor Srinath Sridhar</i>	
Zhejiang University , B.Eng. in Robotics Engineering, Chu Kochen Honor College	September 2020 – June 2024
• GPA: 3.99/4.0	
• Advisors: <i>Professor Qixing Huang</i>	
Hong Kong University of Science and Technology , Exchange Student	January 2023 – June 2023
• Dean's List	
• Advisors: <i>Professor Qifeng Chen & Professor Chenyang Lei</i>	

Research Interests

My research interest lies in broad aspects of Computer Vision, Computer Graphics and Machine Learning topics, more specifically 3D spatiotemporal visual perception, understanding and reasoning of human physical interactions with the world. I am also interested in understanding generative models (diffusion, flow model...) from both theoretical and application perspectives.

Publications

DyTact: Capturing Dynamic Contacts in Hand-Object Manipulation	May 2025
<i>Xiaoyan Cong</i> , Angela Xing Chandradeep Pokhariya, Rao Fu, Srinath Sridhar	
<i>Under Review</i> [arxiv: 2506.03103]	
GenHSI: Controllable Generation of Human-Scene Interaction Videos	May 2025
Zekun Li, Rui Zhou, Rahul Sajnani <i>Xiaoyan Cong</i> , Daniel Ritchie, Srinath Sridhar	
<i>Under Review</i> [arxiv: 2506.19840]	
Art3D: Training-Free 3D Generation from Flat-Colored Illustration	April 2025
<i>Xiaoyan Cong</i> , Jiayi Shen Zekun Li, Rao Fu, Tao Lu, Srinath Sridhar	
<i>IEEE Conference on Computer Vision and Pattern Recognition (CVPR) AI4CC Workshop, 2025, Oral</i> [arxiv: 2504.10466]	
OscillationInversion: Understand the structure of Large Flow Model through the Lens of Inversion Method	October 2024
Yan Zheng, Zhenxiao Liang, <i>Xiaoyan Cong</i> , Yi Yang Lanqing Guo, Yuehao Wang, Peihao Wang, Zhangyang Wang	
<i>Under Review</i> [arxiv: 2411.11135]	
Automatic Controllable Colorization by Imagination	June 2024
<i>Xiaoyan Cong</i> , Yue Wu, Qifeng Chen, Chenyang Lei	
<i>IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024</i> [arxiv: 2404.05661]	
4DRecons: 4D Neural Implicit Deformable Objects Reconstruction from a single RGB-D Camera with Geometrical and Topological Regularizations	2024
<i>Xiaoyan Cong</i> , Haitao Yang, Liyan Chen, Kaifeng Zhang, Li Yi, Chandrajit Bajaj, Qixing Huang	
<i>Under Review</i> [arxiv: 2406.10167]	

Selected Awards and Honors

Excellence Scholarship , by Chu Kochen Honors College, Zhejiang University, Top 1%	2022
Chinese National Scholarship , by Ministry of Education of the People's Republic of China, Top 0.2%	2021

Chunhui Scholarship , by College of Control Science and Engineering, Zhejiang University, Top 1%	2023
Zhejiang Provincial Government Scholarship , Top 2%	2020 - 2022
First-prize Scholarship of Zhejiang University , Top 2%	2020 - 2022

Computer and Language Skills

Programming Languages: Proficient in C/C++, Python (Pytorch), MATLAB.

Technical Skills: Linux/Windows, MeshLab, Blender, SolidWorks, CoppeliaSim, Multisim.

Language: Mandarin (native), English (fluent).